

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

2. A directed graph $G = (V, E)$ of 6 vertices $V = \{0, 1, 2, 3, 4, 5\}$ has edges $E = \{(0, 3), (5, 0), (4, 5), (5, 1), (5, 3), (0, 1), (4, 0), (3, 1), (2, 1), (2, 4), (2, 3), (2, 5)\}$.
- (a) Apply the topological sorting algorithm to G . Show all steps.
 - (b) What is a worst case input for the topological sort algorithm? Justify.

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Answer:

2. (a) In each step we look for a source and remove that source and the edges connected to the source.
- i. the source : 2, $E = \{(0, 3), (5, 0), (4, 5), (5, 1), (5, 3), (0, 1), (4, 0), (3, 1)\}$.
 - ii. the source : 4, $E = \{(0, 3), (5, 0), (5, 1), (5, 3), (0, 1), (3, 1)\}$.
 - iii. the source : 5, $E = \{(0, 3), (0, 1), (3, 1)\}$.
 - iv. the source : 0, $E = \{(3, 1)\}$.
 - v. the source : 3, $E = \emptyset$.
 - vi. the source : 1
- The vertices in sorted order: 2, 4, 5, 0, 3, 1 and the corresponding sorted edges: (2, 1), (2, 4), (2, 3), (2, 5), (4, 5), (4, 0), (5, 0), (5, 1), (5, 3), (0, 3), (0, 1), (3, 1).
- (b) We look for a source for the first empty edge list, running over all vertices. This loop stops at the first empty edge list.
- If the next source happens to be the last vertex each time, then we have to traverse all remaining vertices. This is a worst case input.